10. *(a)* Halogenoalkanes can be hydrolysed using water in a similar way to using aqueous sodium hydroxide.

A student carried out an experiment to investigate the rate of reaction for the hydrolysis of halogenoalkanes using water. The student used aqueous ethanol to dissolve the halogenoalkane and then added a few drops of aqueous silver nitrate. He timed how long it took to produce a precipitate. He obtained the results shown in the table.

Halogenoalkane	Time / s
1-chloropropane, C_3H_7Cl	300
1-bromopropane, C_3H_7Br	90
1-iodopropane, C_3H_7I	15

The student tried to explain these results and he looked on the internet to find the following data.

Bond	Bond enthalpy / $kJ\,mol^{-1}$
C—H	413
C—C	348
C—F	485
C—Cl	328
C—Br	276
C—I	240

Element	Electronegativity
chlorine	3.16
bromine	2.96
iodine	2.66
carbon	2.55

Examiner only

(i) Use both sets of data to explain the student's results. Include an equation to show the hydrolysis reaction between a halogenoalkane and aqueous sodium hydroxide and name this type of reaction mechanism. [6 QER]

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(ii) Write an **ionic** equation, including state symbols, for a reaction that produces a silver halide precipitate. [1]

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(iii) Suggest a practical method by which the student could have obtained these results. [2]

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(iv) Suggest the difference that the student would have observed in his experiments if he had not added ethanol to the water before adding the halogenoalkane. [1]

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Examiner only

(b) Chlorofluorocarbons, CFCs, were historically used for a variety of commercial and domestic purposes but nowadays their use is very restricted.

(i) Outline the adverse environmental effects of the use of CFCs.

You do not need to include any equations for the reactions involved. [4]

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(ii) Use relevant data in part *(a)* to explain why hydrofluorocarbons, HFCs, have replaced CFCs in many of their uses. [2]

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